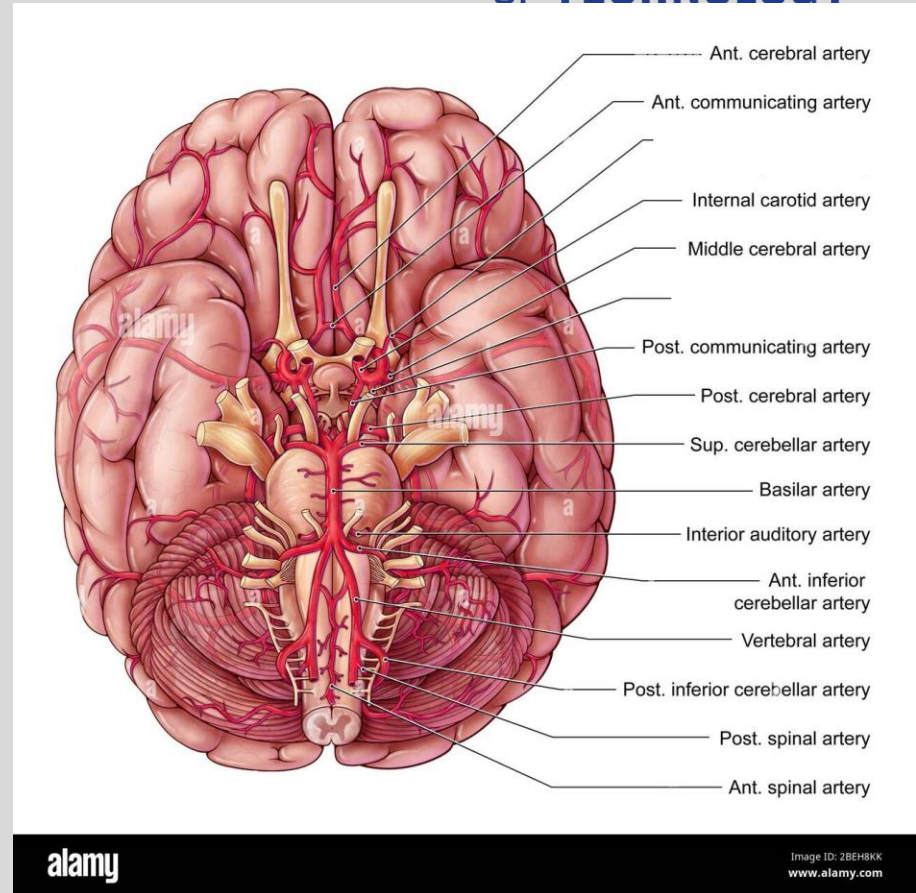


Session Objectives

1. Identify the four major vessels that supply blood to the brain.
2. Differentiate between the anterior and posterior circulations of the brain, including their components.
3. Describe the formation and function of the Circle of Willis.
4. Locate and identify the regions of the brain supplied by each major artery using gross anatomical landmarks.
5. Explain the structures of the cerebral venous system and their roles in venous drainage.
6. Classify and differentiate the types of intracranial hemorrhages based on their origin of bleed and the compartments of the brain they affect.

Arteries you should know by the end of this lecture

- CCA, common carotid artery
- ECA, external carotid artery
- ICA, internal carotid artery
- MCA, middle cerebral artery
- ACA, anterior cerebral artery
- ACom, anterior communicating artery
- PCom, posterior communicating artery
- VA, vertebral artery
- BA, basilar artery
- PCA, posterior cerebral artery
- AICA, anterior inferior cerebellar artery
- PICA, posterior inferior cerebellar artery
- SCA, superior cerebellar artery
- PA, pontine artery
- ASA, anterior spinal artery
- PSA, posterior spinal artery



Arteries you should know by the end of this lecture

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- CCA, common carotid artery
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- PICA, posterior inferior cerebellar artery
- SCA, superior cerebellar artery
- PA, pontine artery
- ASA, anterior spinal artery
- PSA, posterior spinal artery

1. **Parent vessel** → where does it come from?
2. **Destination** → is it giving another branch or supplying a brain area?
3. **Adjacent structures** → what other neurovascular structures are relative to it?
4. **Blood supply** → what part of the brain does it supply blood to?

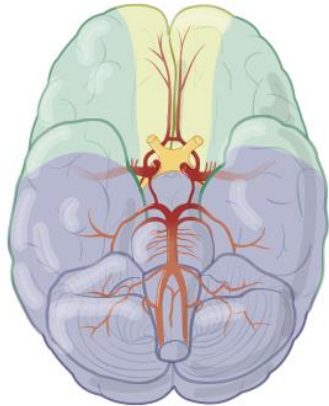
Master this by the end

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Anterior
circulation
ACA
ICA
MCA

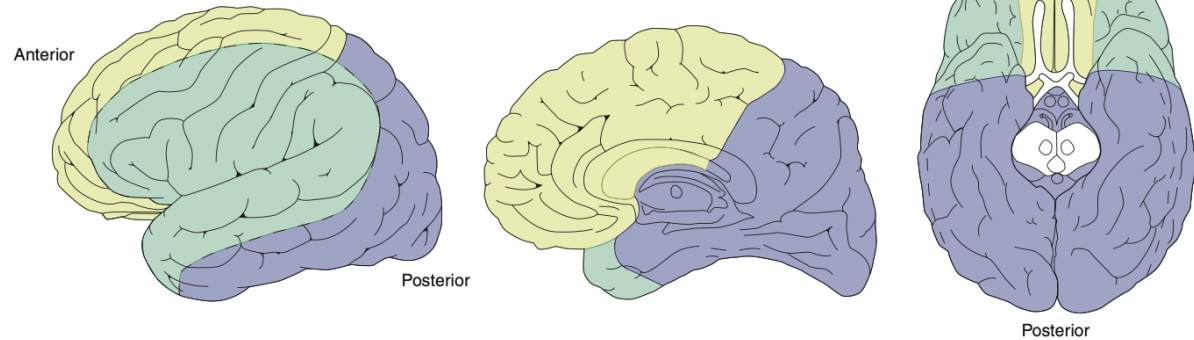
Posterior
circulation



INFERIOR VIEW

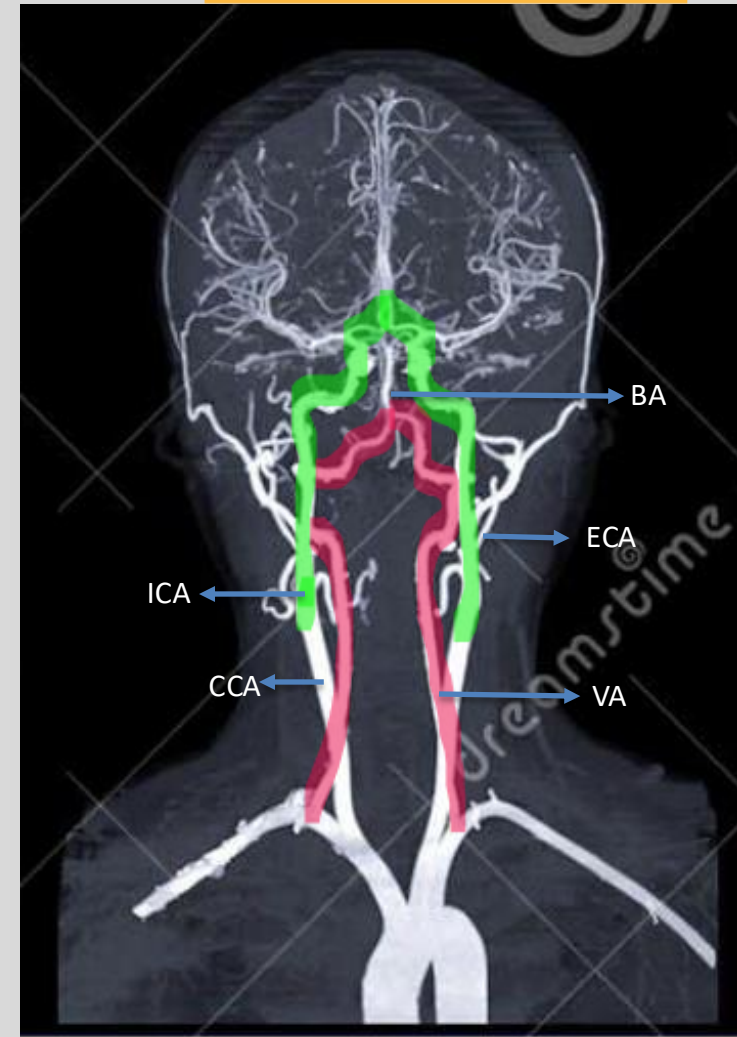
Cerebral arteries—cortical distribution

Anterior cerebral artery (supplies anteromedial surface)
Middle cerebral artery (supplies lateral surface)
Posterior cerebral artery (supplies posterior and inferior surfaces)



The four major arteries

- **2 Internal carotid arteries (ICA)** → anterior circulation
 - ~80% of blood supply
 - Telencephalon and diencephalon
- **2 Vertebral arteries (VA)** → posterior circulation
 - ~20%
 - Diencephalon
 - Brainstem (midbrain, pons, medulla), cerebellum
- Anterior and posterior circulations connected by **posterior communicating arteries (PCom)**



The four major arteries

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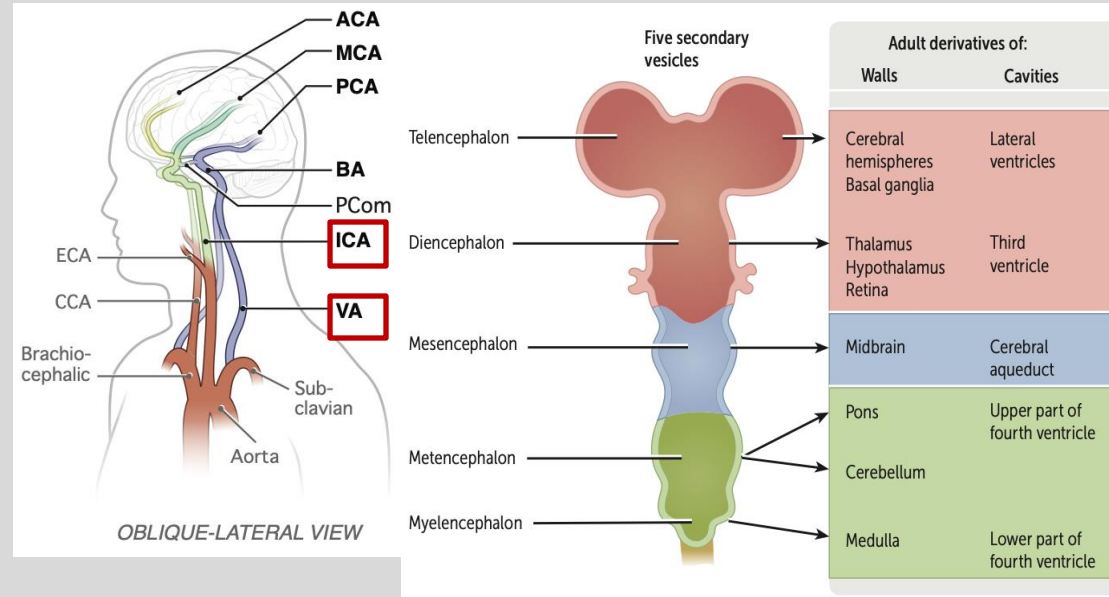
2 Internal carotid arteries (ICA) →
anterior circulation

Telencephalon and diencephalon

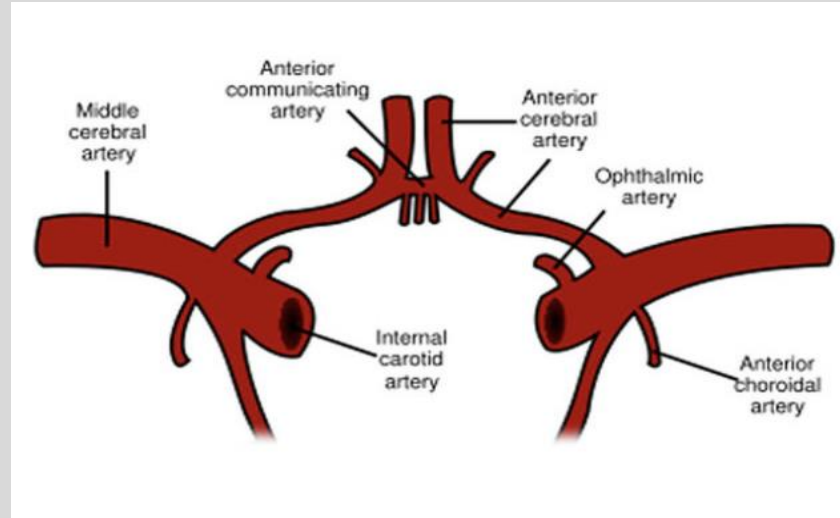
2 Vertebral arteries (VA) → posterior
circulation

Diencephalon

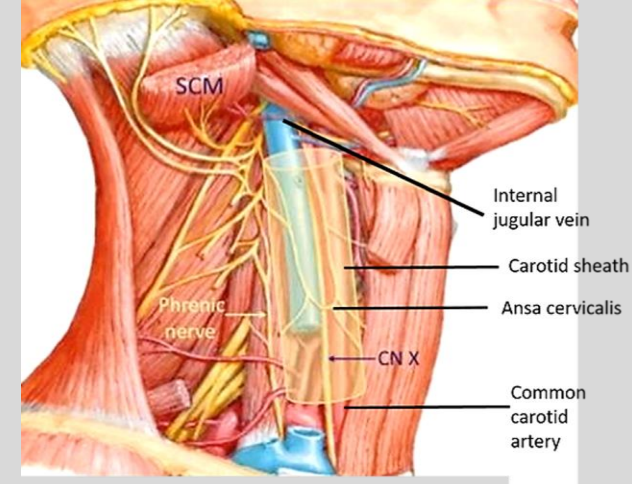
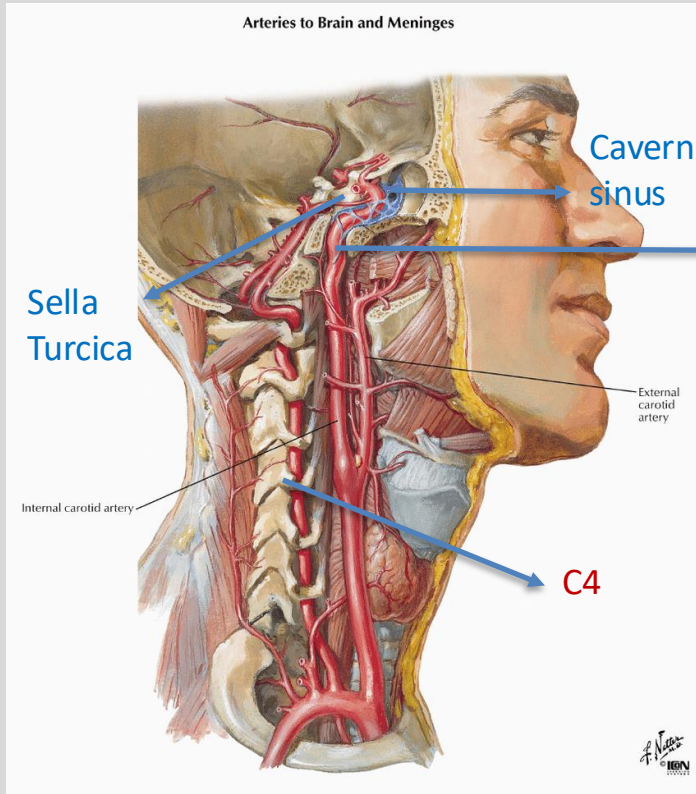
Brainstem (midbrain, pons,
medulla), cerebellum



ANTERIOR CIRCULATION



Internal Carotid Artery (ICA)



- ICA Travels in the **carotid sheath** in the neck and enters the skull through the '**carotid canal**' in the petrous part of temporal bone.
- Once inside the skull → **intracranial ICA**
- It travels through the **cavernous sinus** next to Sella turcica.
- Once it traverses the dura → **intradural ICA**
- Intradural ICA branches: (**OPAM**)
 1. Ophthalmic artery
 2. Posterior Communicating artery
 3. Anterior Cerebral artery
 4. Middle Cerebral artery

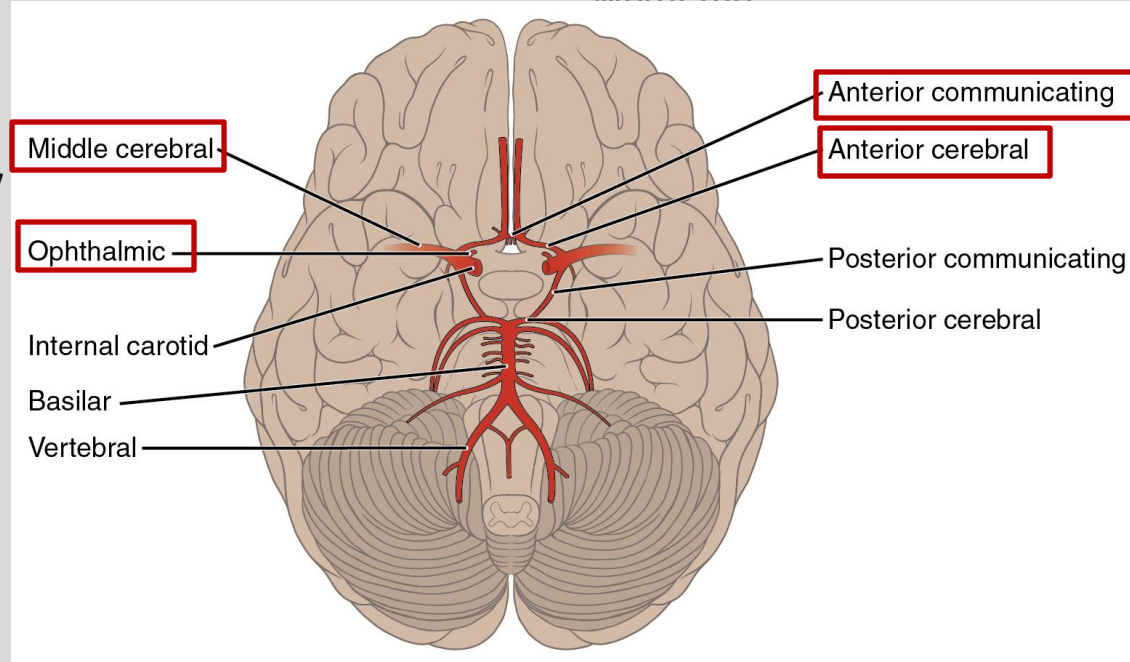
Intradural ICA Branches

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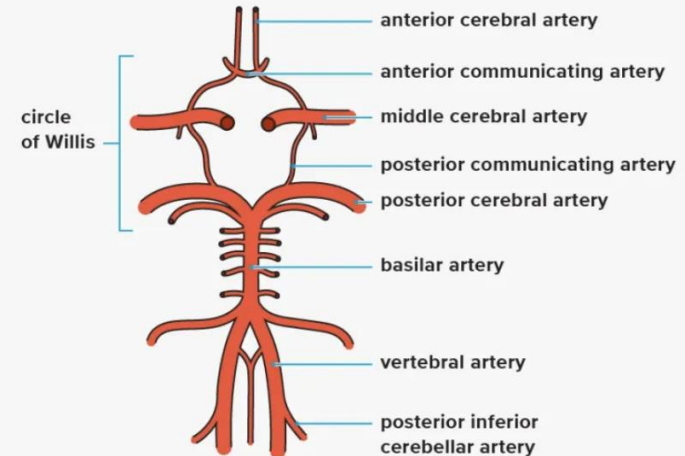
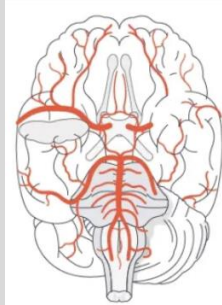
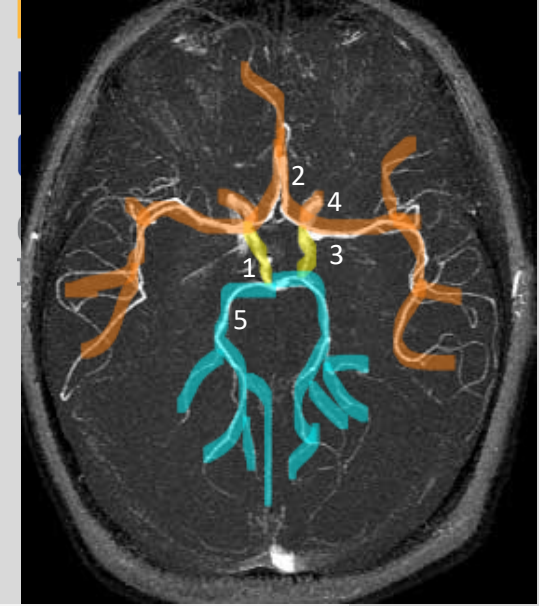
OPAM:

1. **O**phthalmic Artery
2. **P**osterior Communicating Artery
3. **A**nterior Cerebral Artery
4. **M**iddle Cerebral Artery



OPAM: Posterior Communicating Artery (PCom)

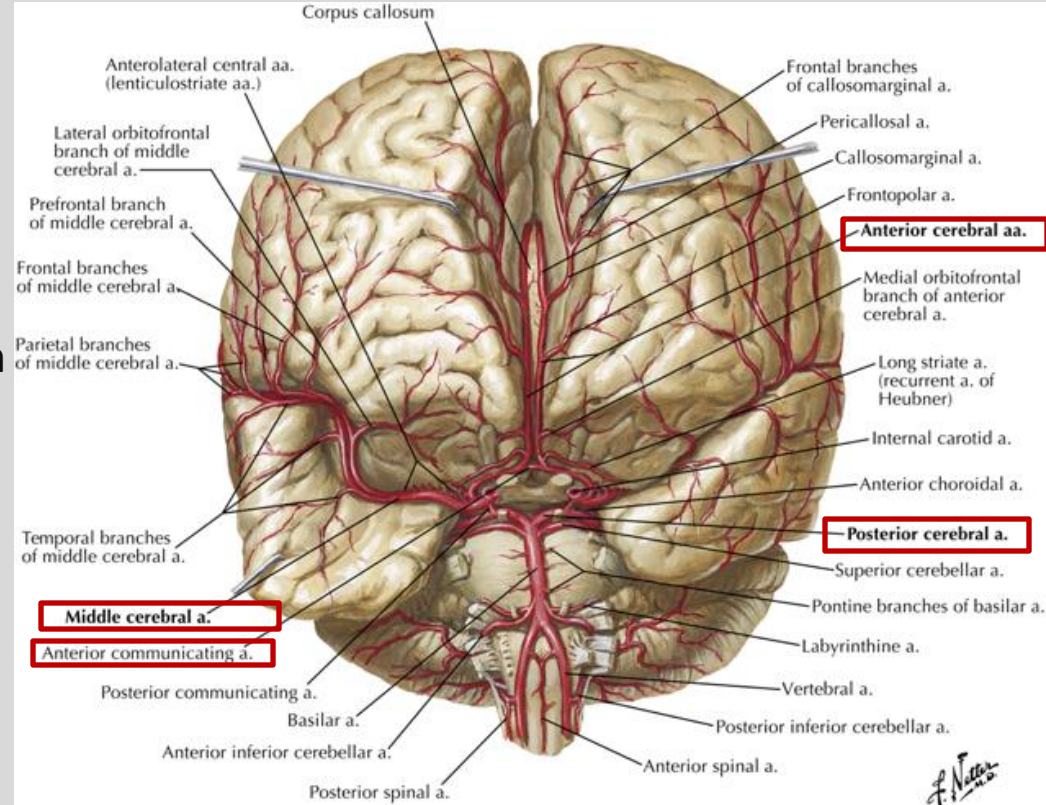
- As the name suggests, PCom communicates the anterior and posterior circulation that forms the circle of Willis.
- Patency of PCom is crucial for adequate circulation of the circle of Willis.



OPAM: Anterior Cerebral Artery (ACA)

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- ICA bifurcates into anterior cerebral artery (ACA) medially and middle cerebral (MCA) artery laterally
- Bifurcation happens under the frontal lobe, in the medial border of the sylvian fissure.
- The left and right ACA communicate with each other through the **anterior communicating artery (ACom)**.
- The ACA then travels in the **interhemispheric** or the longitudinal fissure to **supply blood to the medial and anterior portions of the frontal, temporal and parietal lobes of the cerebrum**.



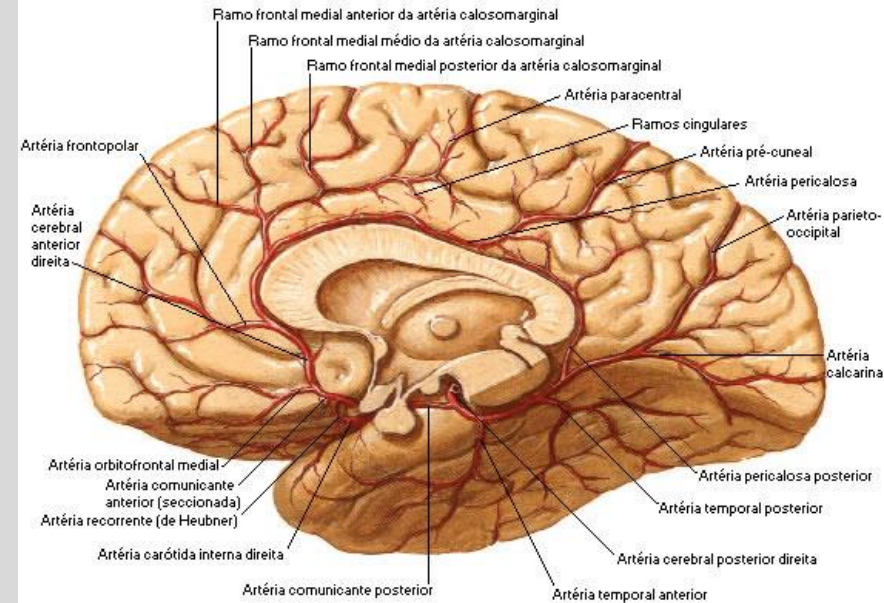
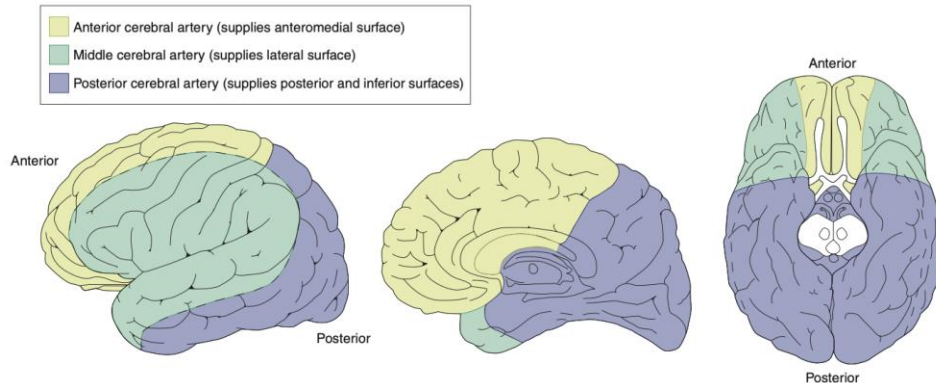
OPAM: Anterior Cerebral Artery (ACA)

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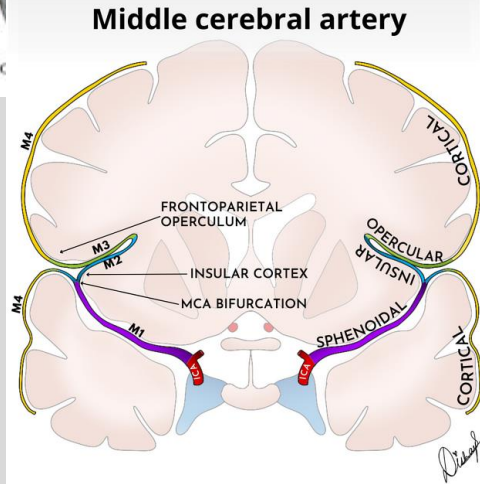
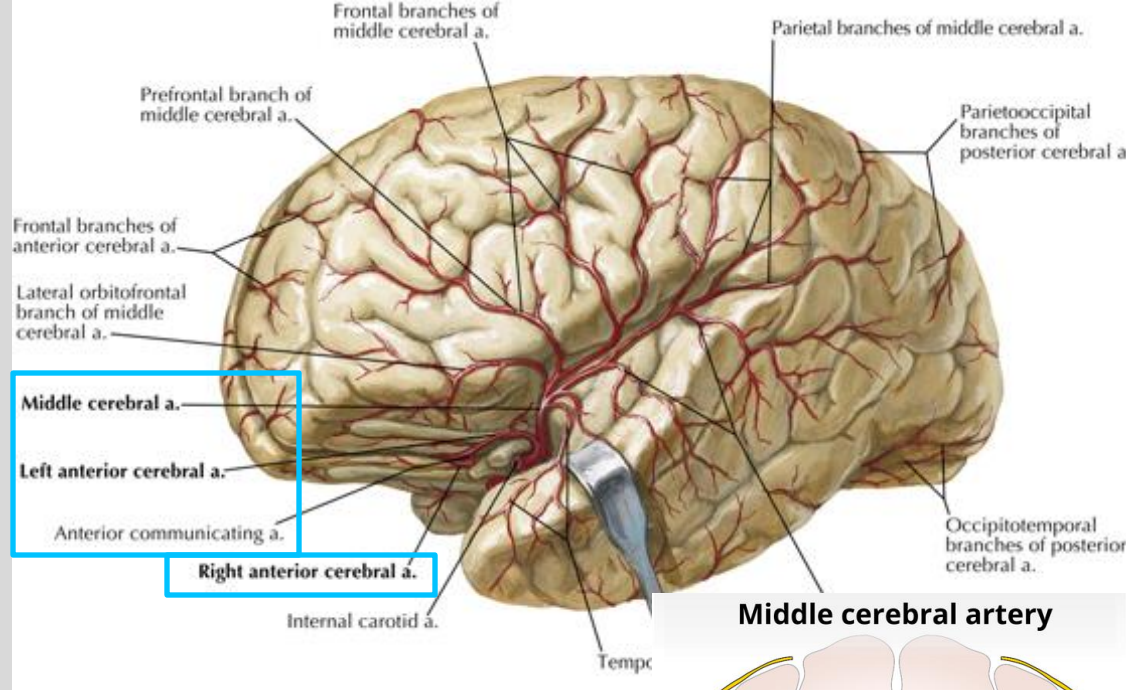
- Travels antero-medially above the **optic nerve** and **chiasm**.
- **Supply blood to the medial and anterior portions of the frontal and parietal lobes of the cerebrum.**

Cerebral arteries—cortical distribution



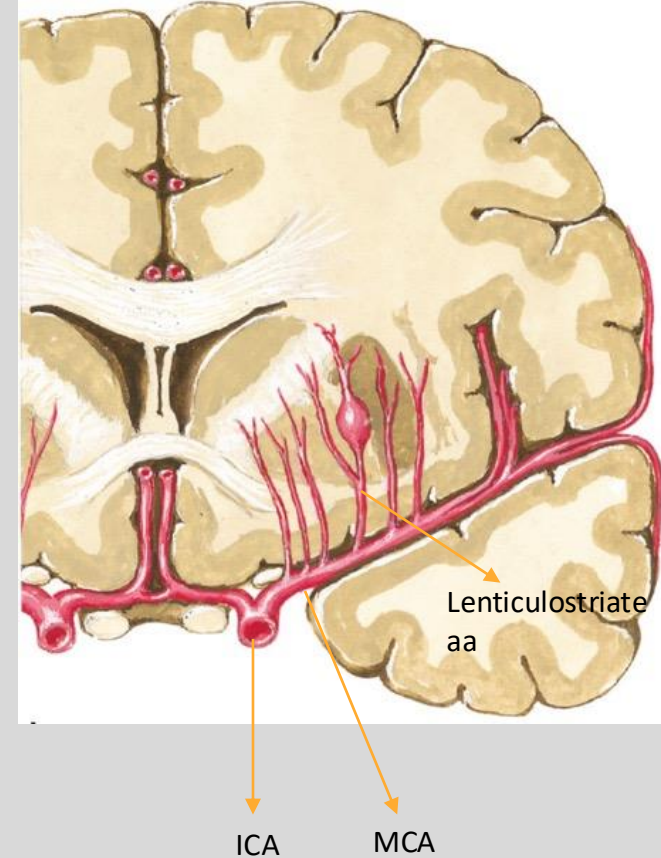
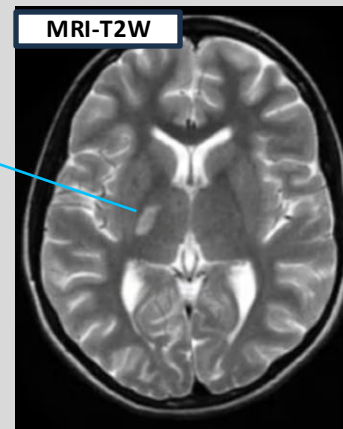
OPAM: Middle Cerebral Artery (MCA)

- ICA bifurcates into **anterior cerebral artery (ACA)** medially and **middle cerebral (MCA)** artery laterally
- The middle cerebral artery is **the larger terminal branch** of the internal carotid artery.
- Bifurcations happens on the medial border of the sylvian fissure.
- The MCA then follows the borders of the **sylvian fissure laterally, wraps around the insular cortex** and emerge on the cortical surface of the brain **supplying much of the lateral parts of the frontal, temporal and parietal lobe.**



OPAM: Middle Cerebral Artery (MCA)

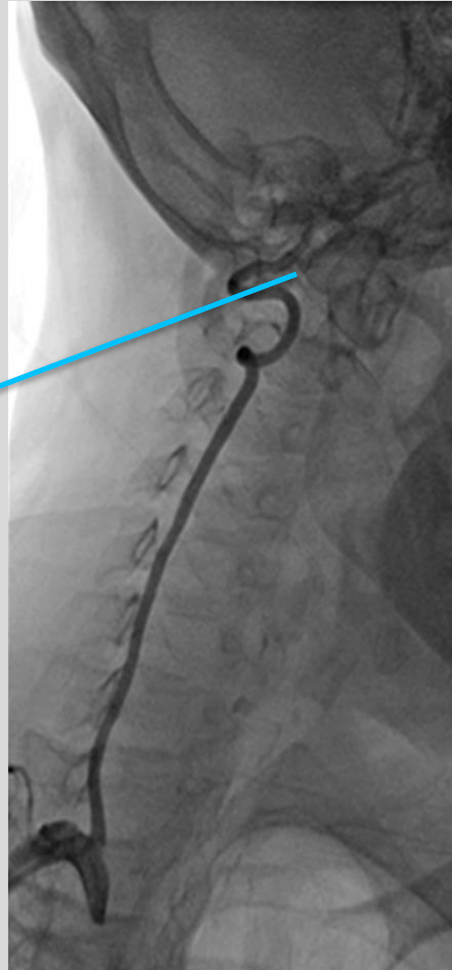
- An important deep branch of MCA is the tiny **lenticulostriate arteries** that supply the **basal ganglia**.
- These arteries are vulnerable to the formation of small aneurysms, especially in the setting of **chronic, uncontrolled hypertension**. When these aneurysms rupture, they can lead to hemorrhages known as **Charcot-Bouchard aneurysm**.
- Lenticulostriate arteries are also **susceptible to ischemic strokes** given their size and are known as **lacunar strokes**, named for the characteristic “lake-like” appearance of the resulting cavities in the brain tissue on MRI of the brain.



More on this in the cerebrovascular pathology lecture later..

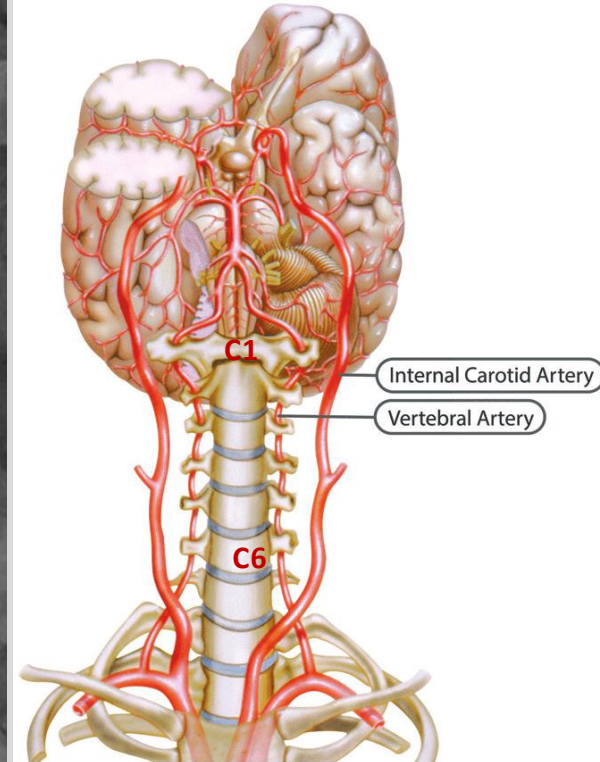
Vertebral Artery (VA)

- **Subclavian artery → vertebral artery (VA)**
- Travel through the **transverse foramen of C6-C1 vertebrae**
- **Enter the cranium via the foramen magnum, lateral and ventral to the brainstem.**
- The **left and right vertebral arteries fuse** to become the **basilar artery (BA)** below the level of pons.



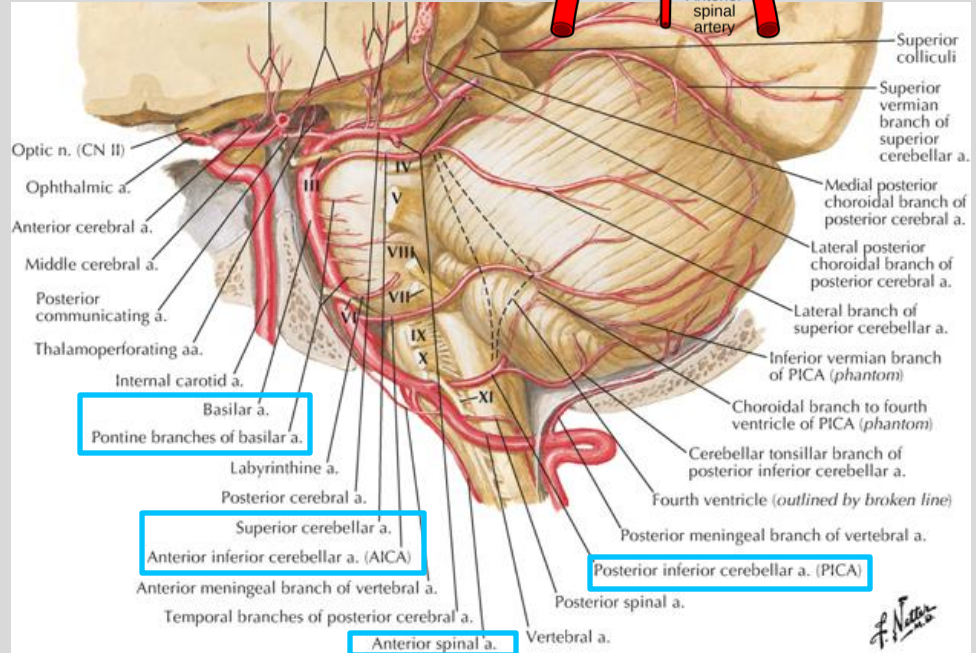
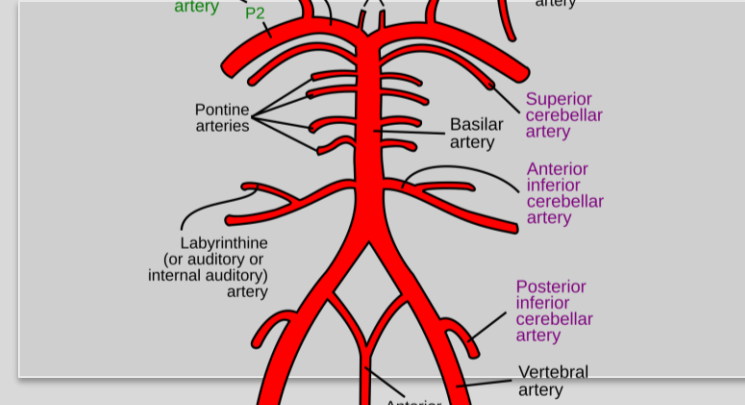
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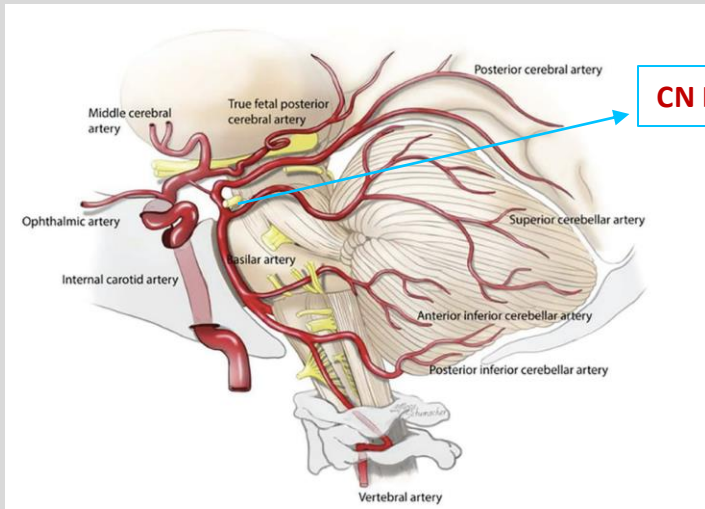
Vertebrobasilar system

- The vertebral arteries and its branches + basilar arteries and its branches → **vertebrobasilar system**: makes up the infratentorial portion of the posterior circulation.
- Vertebral artery branches:
 - Posterior inferior cerebellar artery (PICA)**
 - Supply: inferior and medial cerebellum
 - Anterior spinal artery (ASA)**
 - Supply: anterior spinal cord
- Basilar artery branches:
 - Anterior inferior cerebellar arteries (AICA)**
 - Supply: anterior and lateral cerebellum
 - Pontine arteries (PA)**
 - Supply: basilar pons
 - Superior cerebellar arteries (SCA)**
 - Supply: superior- medial and superolateral cerebellum



Posterior Cerebral Artery (PCA)

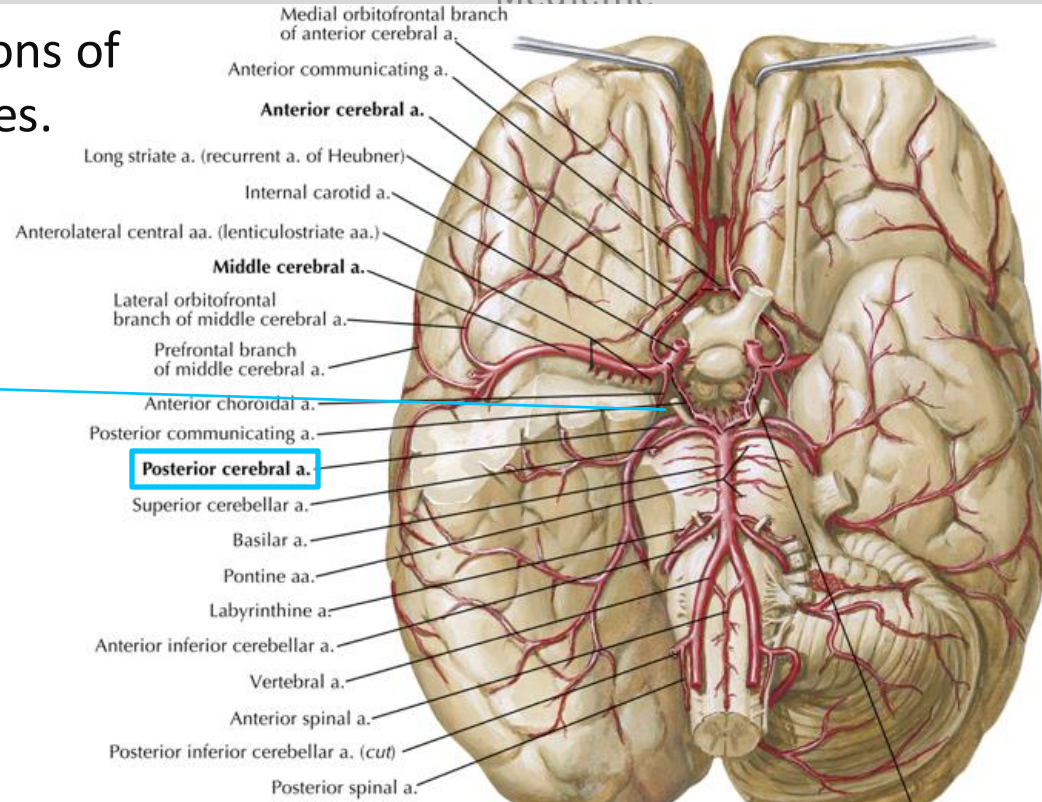
- **Posterior cerebral artery (PCA)** is the terminal branch of basilar artery.
- **Supplies:** inferior and medial portions of the cerebrum and the occipital lobes.
Above the tentorium
- **CN III travels b/w PCA and SCA**



CN III

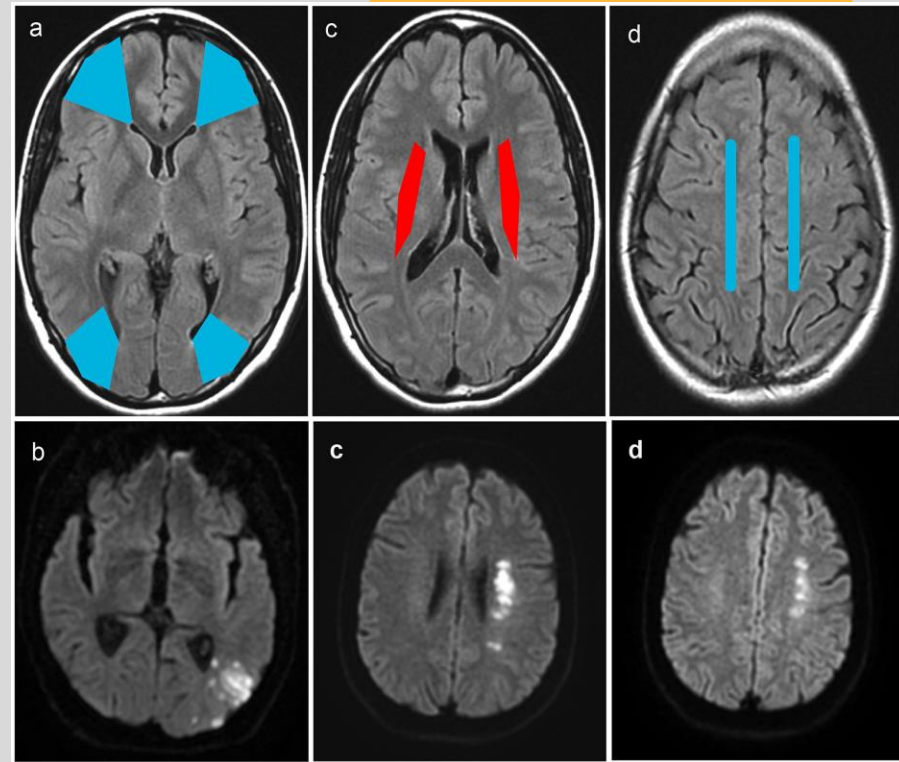
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Watershed Area

- Regions in the brain situated at the **border zones** between major cerebral arteries, particularly the anterior, middle, and posterior cerebral arteries.
- They are **prone to infarction** during **global hypoperfusion**.
- Cortical Watershed Zones:**
 - Outer cortical areas where **ACA, MCA, and PCA territories meet**.
- Subcortical Watershed Zones:**
 - Deep brain regions, typically between the deep vascular territories of the MCA and ACA

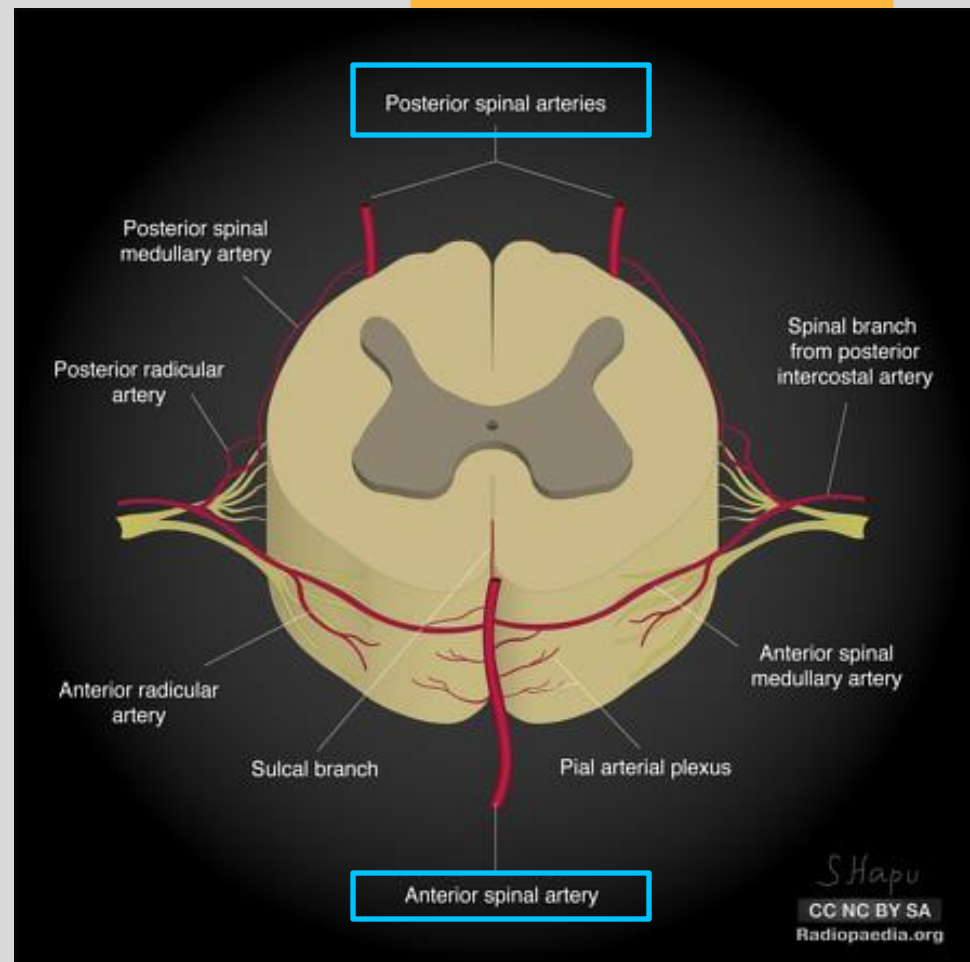


Watershed territories, on MRI (FLAIR); in blue: cortical watershed, in red: internal watershed; (a) frontal WSI: between the anterior and the middle cerebral arterial territories, (a, b) occipital WSI: at the boundaries of the middle and the posterior arterial territories; (c) internal WSI: between territories of superficial and deep perforators of the middle cerebral artery; (d) paramedian WSI: between the anterior and the middle cerebral arterial territories.

More on this in the cerebrovascular pathology lecture later..

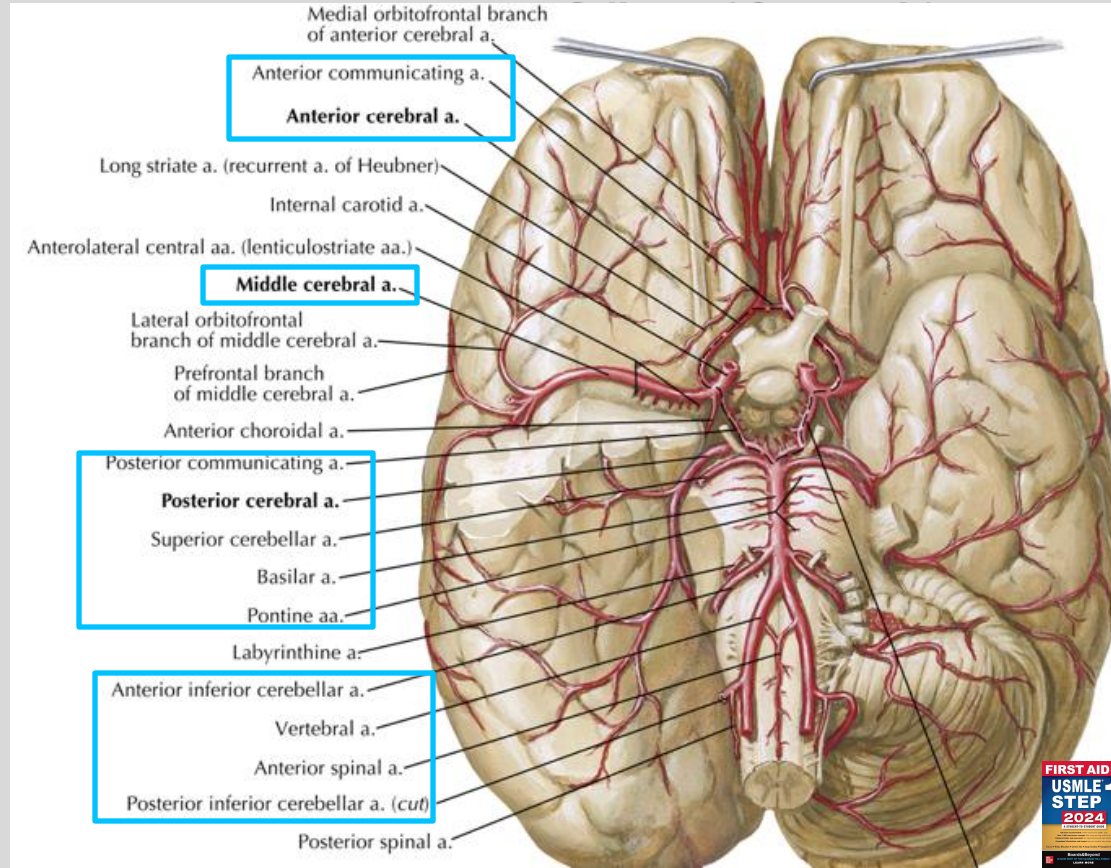
Spinal arteries

- Vertebral artery → **1 Anterior spinal artery (ASA)**
 - Supply: anterior spinal cord (mostly motor)
- Posterior inferior cerebellar artery (PICA) → **2 Posterior spinal artery (PSA)**
 - Supply: posterior spinal cord (mostly sensory)



Circle of Willis (CoW)

- A circular network of arteries at the **base of the brain** that provides **collateral circulation**.
- Formed by the **joining of the internal carotid arteries** (anterior circulation) and the vertebral arteries (posterior circulation) via **PCom**.
- Acts as a safeguard**: if one major artery is blocked, the Circle of Willis can potentially provide alternate routes for blood flow, reducing the risk of ischemia.
- Common site for the development of saccular (berry) aneurysms**, particularly at branch points such as the **ACom**.
- Anatomical variations** are common in the Circle of Willis, with complete and functional circles present in only about 30-50% of individuals.

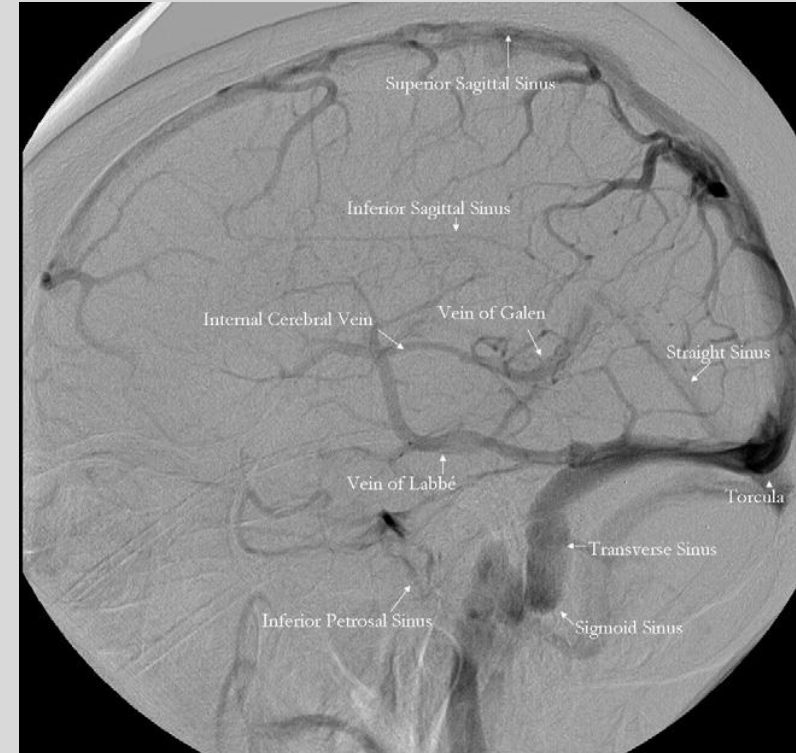
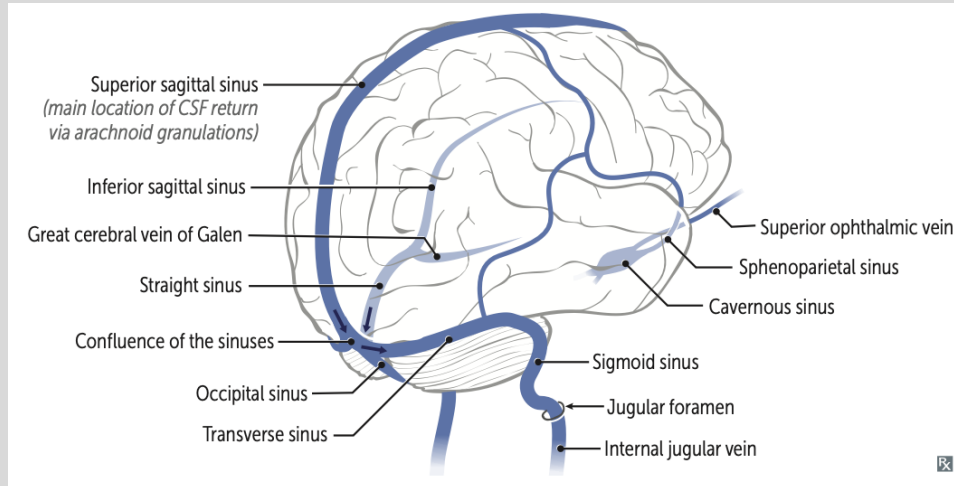


CEREBRAL VENOUS SYSTEM

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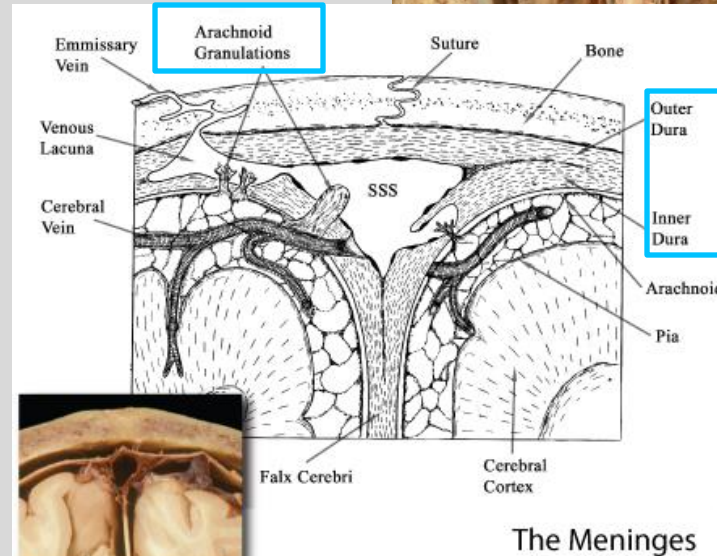
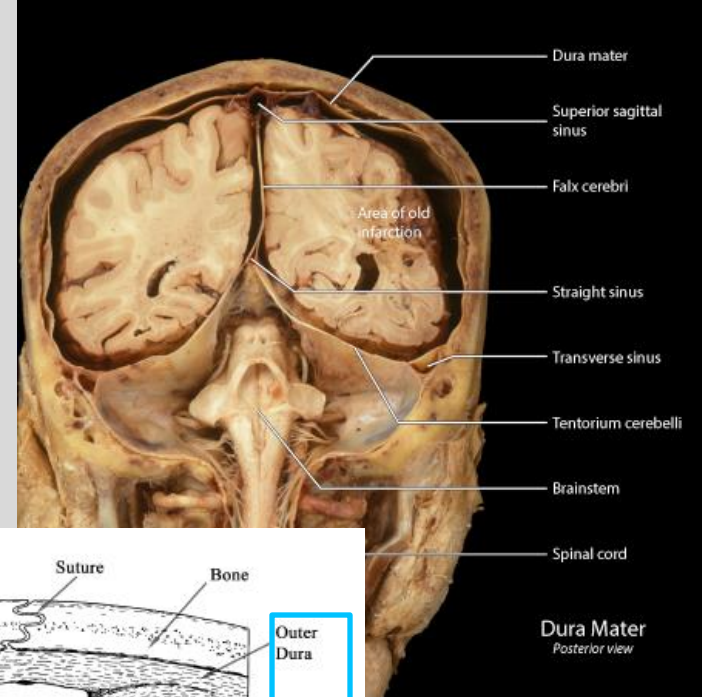
- Cerebral sinuses
- Deep cerebral veins
- Superficial cerebral veins



Cerebral Sinuses

Appropriately called **dural sinuses**.

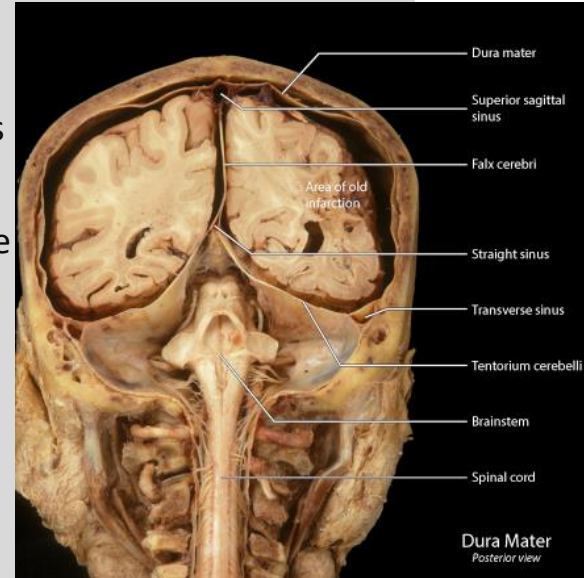
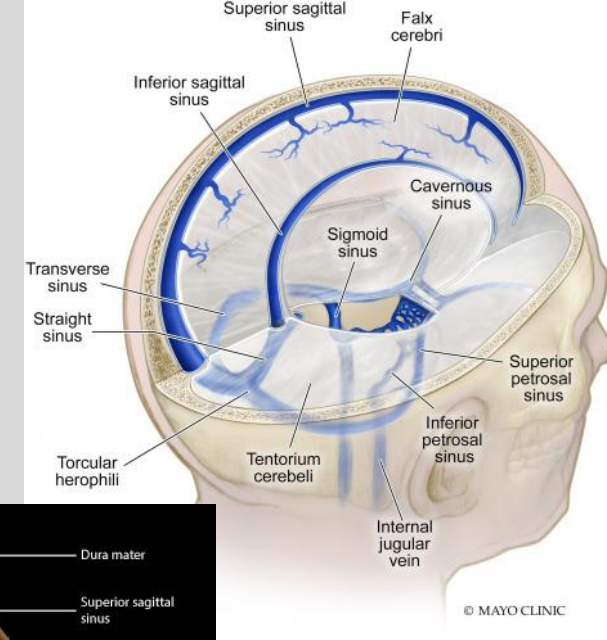
- **Channels between layers of dura mater that drain deoxygenated blood and cerebrospinal fluid (CSF) from the brain.**
- While referred to as venous structures, they are **not veins**
 - Lack the 3 tunica layers
 - Lack valves
 - Walls are simply dural mater
- Made by the **infoldings of falx cerebri and tentorium cerebelli**
- Have small **outpouching** on their surface called **arachnoid granulations**:
 - Drain CSF from subarachnoid space to venous system



Cerebral Sinuses

Major dural sinuses:

- **Superior sagittal sinus:** Drains blood from the superficial veins and flows into the confluence of sinuses.
- **Inferior sagittal sinus:** Joins the straight sinus, draining deep brain structures.
- **Straight sinus:** Connects the inferior sagittal sinus and the great vein of Galen.
- **Cavernous sinus:** unique dural venous structure drains orbit and face.
- **Transverse sinuses:** Receive blood from the confluence of sinuses and drain into the sigmoid sinuses.
- **Sigmoid sinuses:** Drain into the internal jugular veins (IJVs), completing the venous outflow.
- **Confluence of sinuses/ Torcular herophili :** Meeting point of superior sagittal, straight, and transverse sinuses.



A

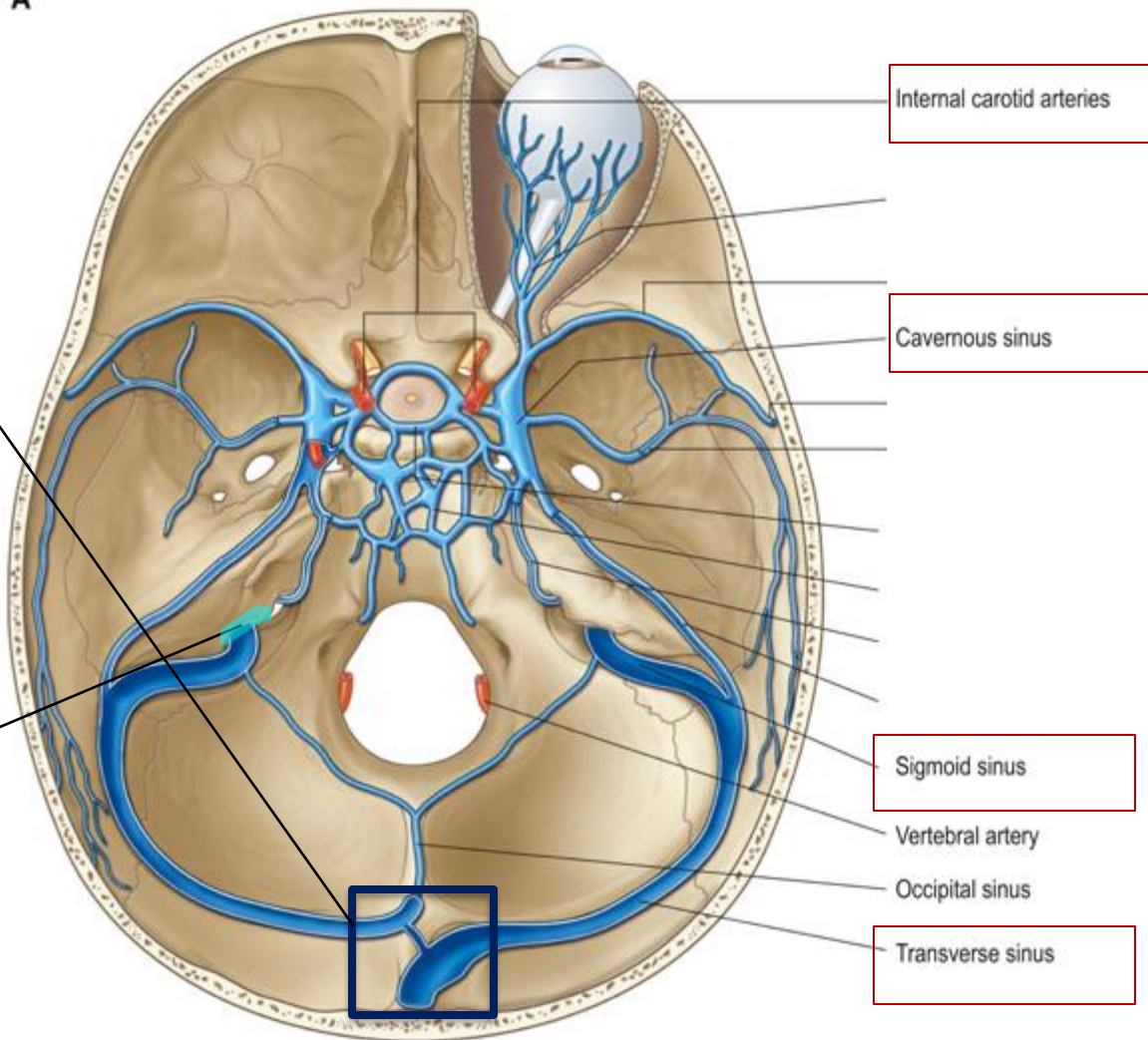
Cerebral Sinuses

Transverse Sinus

- Originates at the **confluence of sinuses in the posterior skull**, lies laterally in the groove of occipital bone.
- **Drains blood from the superior sagittal sinus, straight sinus, and occipital sinus.**

Sigmoid Sinus

- Continuation of the transverse sinus at the ‘**transverse-sigmoid junction**’
- Courses through the skull base to join the **internal jugular vein** at the **jugular foramen**.
- **Proximity to cranial nerves and the jugular foramen**; important in neurosurgical approaches.



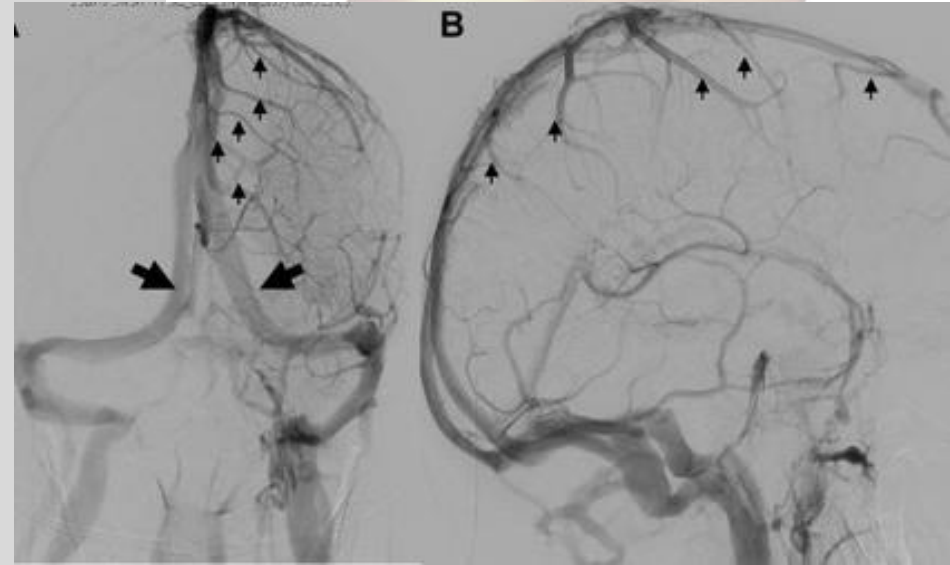
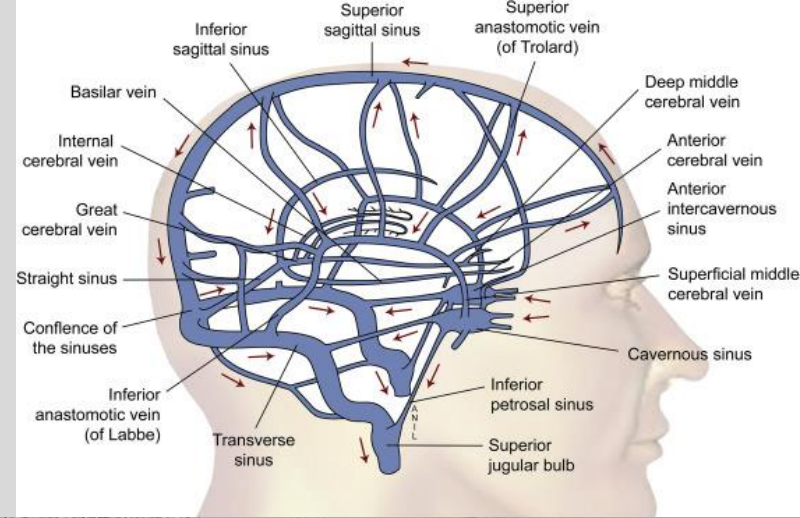
Direction of Drainage

Superficial Venous System: Drains the cerebral cortex into the **superior sagittal sinus** and **transverse sinuses**

Deep Venous System: Drains deep brain structures into the **internal cerebral veins** and **straight sinus**

Blood from the superficial and deep venous systems converges at the **confluence of sinuses**

From here, it drains into the **transverse sinuses**, then to the **sigmoid sinuses**, and exits via the **internal jugular veins** through the **jugular foramen**



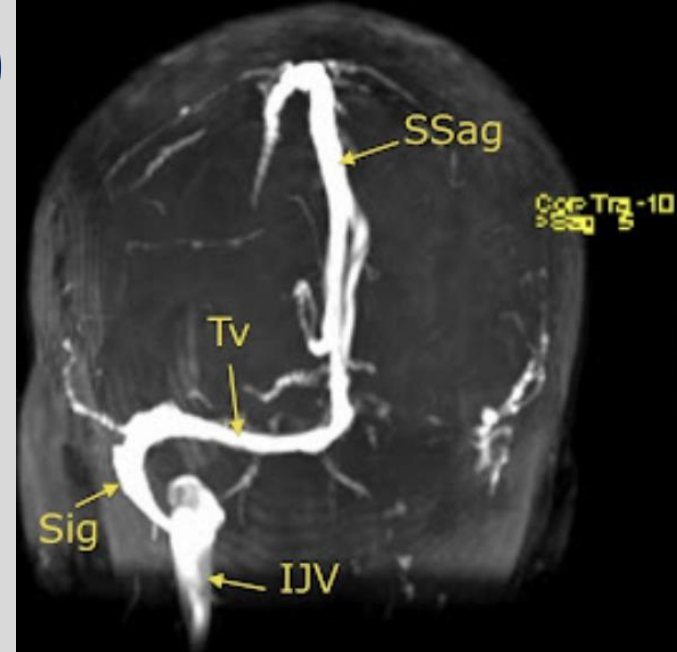
Cerebral Venous Thrombosis (CVT)

CVT is a blood clot in the dural venous sinuses or cerebral veins, blocking venous drainage from the brain. Common sites:

- Superior sagittal sinus
- Transverse sinus
- Sigmoid sinus
- Cortical veins

Risk Factors: Hypercoagulable states (e.g., pregnancy, oral contraceptive use, malignancy), infections (e.g., mastoiditis, sinusitis), head injury, chronic inflammatory diseases (e.g., lupus, IBD)

Clinical Presentation: Headache (often severe and sudden), seizures, focal neurological deficits (e.g., weakness, vision changes), symptoms of increased intracranial pressure (e.g., papilledema, nausea, vomiting).



MRV: Venous thrombosis of left transverse sinus at the junction of confluence of sinuses

More on this in the cerebrovascular pathology lecture later..

Knowledge check:

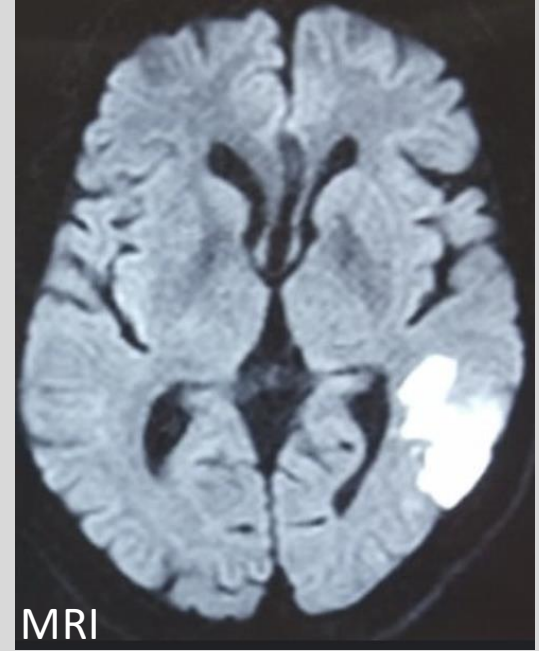
CEREBRAL VENOUS SYSTEM

1. **Parent Vessel**→ Which major veins or sinuses does this vessel drain to?
 - a. Is it part of the superficial or deep venous system?
2. **Destination**→ Where does this venous pathway lead (e.g., confluence of sinuses, internal jugular vein)?
 - a. Does it converge with other veins or directly exit the cranial vault?
3. **Adjacent Structures**→ Which neurovascular structures are located near this sinus or vein?
 - a. Are there any dural folds, cranial nerves, or arterial branches in close proximity?
4. **Drainage Territory**→ What regions of the brain does this sinus or vein drain (e.g., cortex, deep brain structures)?

Clinical question

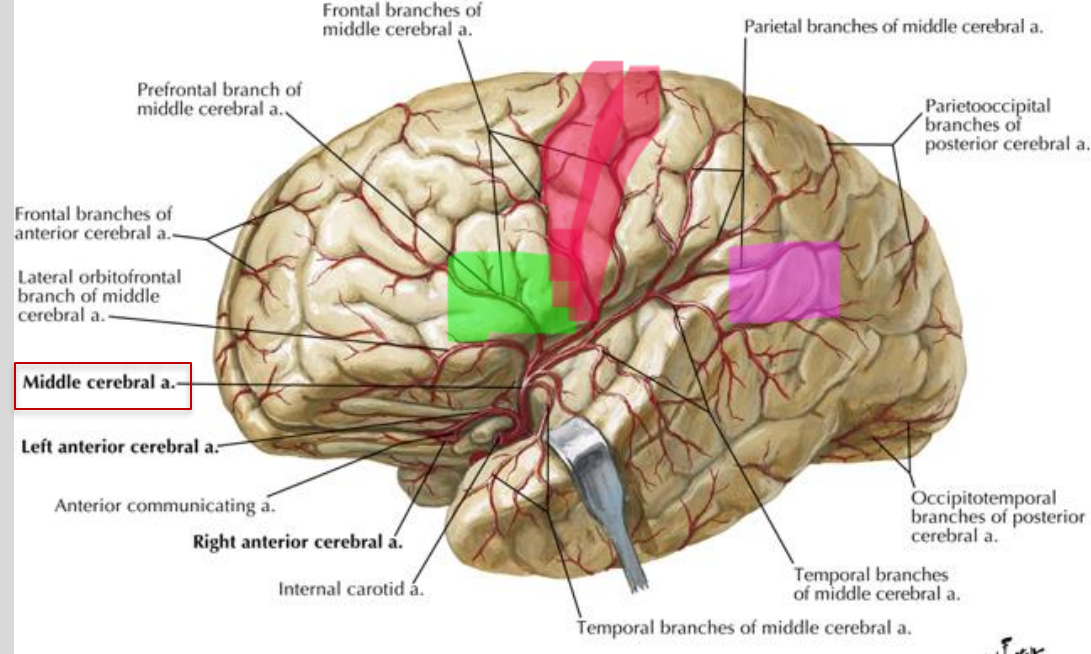
A 65-year-old man is brought to the emergency department with sudden difficulty understanding language. He speaks fluently but uses nonsensical words and does not comprehend spoken or written language. MRI shows an ischemic stroke affecting a specific artery. Which artery is most likely involved in causing these symptoms?

- A. Anterior cerebral artery
- B. Middle cerebral artery
- C. Posterior cerebral artery
- D. Basilar artery
- E. Vertebral artery



Broca's and Wernicke's Area supply

- **Broca's area**, in the opercular and triangular parts of the inferior frontal gyrus.
 - Lesion here results in **expressive aphasia (fluency)**
- **Left precentral gyrus**
 - **primary motor cortex**
- **Wernicke's area**, in the posterior superior temporal gyrus.
 - Lesion here results in **receptive aphasia (understating)**

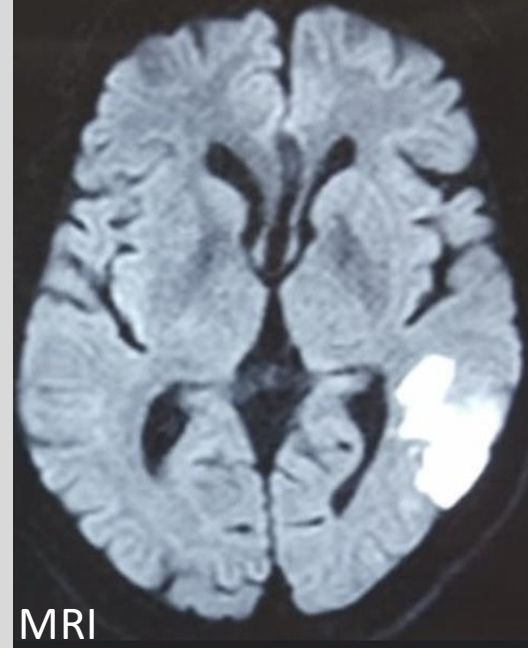


MCA Territory

Clinical question

A 65-year-old man is brought to the emergency department with sudden difficulty **understanding language**. He **speaks fluently but uses nonsensical words** and does not comprehend spoken or written language. MRI shows an **ischemic stroke affecting a specific artery**. Which artery is most likely involved in causing these symptoms?

- A. Anterior cerebral artery
- B. **Middle cerebral artery**
- C. Posterior cerebral artery
- D. Basilar artery
- E. Vertebral artery



The **middle cerebral artery (MCA)** supplies the **lateral aspects of the cerebral cortex**, including **Wernicke's area** in the dominant hemisphere. An infarct in this region can lead to **Wernicke's aphasia**, characterized by **fluent speech without meaningful content and poor language comprehension**. This is commonly associated with a **distal MCA stroke**.

Intraventricular Hematoma

Location: Within the **brain's ventricular system**.

Cause: Can result from **extension of intracerebral hemorrhage** or trauma, commonly seen in **preterm infants**.

Presentation: Variable; may cause symptoms of **hydrocephalus** if blood obstructs CSF flow.

Imaging: Blood in the ventricular spaces on CT.

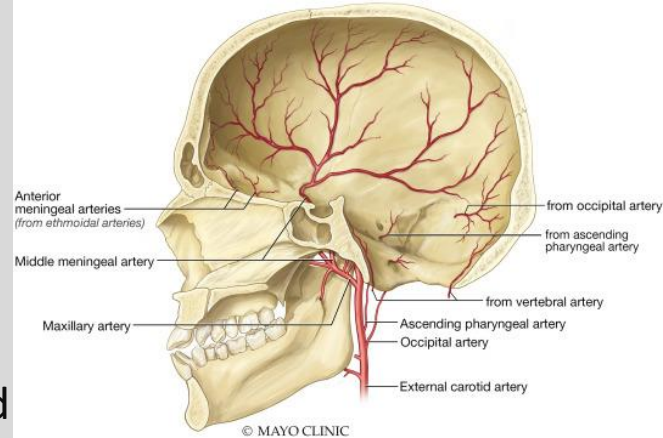


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Clinical question

A 28-year-old man presents to the emergency department following a **motor vehicle accident**. He initially **lost consciousness**, then experienced a **lucid interval** but is now worsening with severe headache, nausea, vomiting, confusion, and one-sided weakness. His vital signs show hypertension and bradycardia, and he begins to have episodes of altered consciousness. A CT scan reveals a **lens-shaped hyperdensity**.



Which vessel is most likely ruptured in this type of bleed?

- A. Middle cerebral artery
- B. Anterior cerebral artery
- C. Bridging dural veins
- D. Middle meningeal artery**
- E. Inferior anastomotic cerebral vein

An epidural hematoma typically results from rupture of the **middle meningeal artery** following trauma, often with an **associated skull fracture**. This leads to a lens-shaped (biconvex) appearance on CT and is commonly characterized by a lucid interval followed by rapid neurological deterioration.

Summary

Arterial Supply:

Anterior Circulation

Supplied by the internal carotid arteries, branching into the anterior and middle cerebral arteries.

Key areas supplied: frontal lobes, parietal lobes, and superior temporal lobes.

Posterior Circulation

Supplied by the vertebral arteries, which form the basilar artery and branch into the posterior cerebral arteries.

Key areas supplied: brainstem, cerebellum, occipital lobes, and inferior temporal lobes.

Venous Drainage:

Superficial System: Includes the superior sagittal sinus and cortical veins draining into dural venous sinuses.

Deep System: Drains deep structures of the brain via the internal cerebral veins into the straight sinus and then to the confluence of sinuses.

Summary

Watershed Areas:

- Border zones between major arteries, particularly vulnerable to ischemia during low blood flow states.

Key Clinical Correlates:

- **Cerebral Venous Thrombosis (CVT):** Obstruction in the dural sinuses or cerebral veins, leading to increased intracranial pressure, venous congestion, and potential infarction.
- **Stroke Syndromes:** Result from blockages in specific arteries; symptoms depend on the artery affected.
- **Intracranial Bleeds:** Epidural, subdural, subarachnoid, and intraparenchymal hemorrhages, each with distinct causes, locations, and imaging features.